DECISION SHEET

Standard(s)- (year and edition):	**Sub clause(s):**	**Sheet n°:**
IEC 60601-1:1988 Ed.2 Am1+Am2	18f)	**DSH-432**
Subject: Protective Impedance versus no-load voltage	**Key words:** Protective, impedance, earth, connection, no-load	**Confirmed by CTL at its 39th meeting, in Cologne**

Question:

EQUIPMENT for example having a rated current of 30 A require a test current of 1.5 times 30 A. With a maximum impedance of 0.2 Ω the voltage drop has to be 9 V (0.2 Ω ∗ 45 A = 9 V). This is in contradiction to the required no-load voltage of 6 V maximum.

Decision:

Measuring the protective earth connection has in fact two reasons. It is to determine impedance and cross-sectional area of protective earth connections. For a measuring current of 25 A both can be done with one measurement. Requiring a measuring current of more than 25 A it shall be splitted up into two measurements. In this case, the impedance shall be determined first, using a voltage not exceeding 6 V.

If cross-sectional area of the protective earth connections cannot be determined as equal to the one for the phase by measurement of the area, then measurement with current shall be from a source with a higher voltage than 6 V.

Explanatory notes:

A circuit to the PROTECTIVE EARTH TERMINAL may have zones of higher impedance, for example due to oxidation of materials. Voltages higher than 6 V prevent detection of such zones because of their ability to flash through. In this case, the impedance shall be determined first, using a voltage not exceeding 6 V.

Using low voltages and low currents has a great impact on the accuracy of the measurement of low impedances. Impedances in the range of 0.1 Ω and 0.2 Ω then require a sophisticated measuring device.

The relation between rated current of the EQUIPMENT and measuring current is because to check cross-sectional areas of protective earth connections. If parts of the construction or printed circuit boards are used for protective earth connections the cross-sectional areas and the ability of carrying short circuit currents are in doubt.